32-bit sign extension → 64 or qwordized

ie rax = qword rax₀

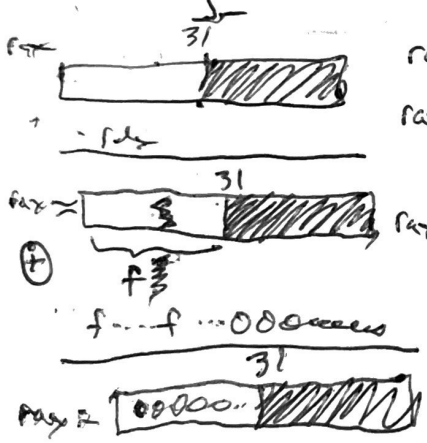
-2^{27}

full 64-bits

```
mov     rdx, 0xffffffff80000000
add     rax, rdx
xor     rax, rdx
```

3+28 = 31

0-5



rdx = 0xffff ffff 8000 0000

rax = rax + rdx

= rax + (ffffffff 8000 0000)

rax = rax ⊕ rdx

= (rax + rdx) ⊕ rdx

qword ~ 64-bits

= (rax ⊕ rdx) + (rdx ⊕ rdx)

= rax 0

ie rax = (int64_t)(int32_t) rax₀